

Title

Rah-e-Ravaan: An AI-Powered Intelligent Travel Planning and Tourism Assistance Platform for Pakistan

Description

Planning trips across Pakistan often requires users to rely on multiple platforms for destinations, transport, accommodations, maps, weather, and travel updates. Rah-e-Ravaan is an AI-powered travel planning platform that simplifies this process through an intelligent conversational assistant available on both web and mobile applications.

The platform leverages conversational AI, Retrieval-Augmented Generation (RAG), recommendation systems, and geospatial intelligence to generate personalized itineraries, provide location-aware recommendations, estimate travel costs, and assist users throughout their journey. It also supports collaborative trip planning, institutional tour management, and real-time travel updates.

Designed with an offline-first mobile architecture, users can download maps and itineraries, and track routes using GPS. The system can also integrate optional GPS tracking devices for group travel monitoring and safety.

Key Outcomes:

- AI-powered conversational travel planner with personalized itinerary generation.
 - RAG-based recommendation engine for intelligent destination discovery.
 - Interactive maps
 - Collaborative group travel and institutional trip management.
 - Real-time travel insights using weather, events, and location data.
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Idea Level

UG

Idea Type

FYP

Tool & Technology

This FYP combines artificial intelligence, mobile and web development, GIS, and cloud computing.

Frontend (Web): React, Vite, Tailwind CSS, Mapbox.

Frontend (Mobile): Flutter, SQLite, GPS & Background Location Services.

Backend: Python/FastAPI, REST APIs, WebSockets.

Database: PostgreSQL, Redis, Qdrant (Vector Database).

AI/ML Module: OpenAI API, LangChain, Sentence Transformers, RAG, Recommendation System.

Cloud & Deployment: Docker, GitHub Actions, Vercel, Railway/Render.

Requirements

Strong knowledge of full-stack development, mobile application development, REST APIs, relational databases, artificial intelligence, natural language processing, GIS technologies, cloud deployment, and real-time systems is recommended.

Floated By

Maajid Maqbool

Preferred Degree

BSCS

Knowledge Domain

Artificial Intelligence & Software Engineering

Knowledge Areas

- Artificial Intelligence
 - Software Engineering
 - Natural Language Processing
 - Recommendation Systems
 - GIS & Location Intelligence
 - Mobile Computing
 - Cloud Computing
 - Database Systems
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School

NUST School of Electrical Engineering and Computer Sciences